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Research Profile

Graduate researcher and NSF Robert Noyce Scholar working at the intersection of quantum computing, compiler-level optimization, and reproducible experimentation. Current research spans quantum network optimization, fault-tolerant routing, and fairness-aware decision workflows. Specifically interested in compiler-assisted precision and resource management for fault-tolerant quantum workloads, as explored in the SPQR project and TAFFO-inspired frameworks for quantum error correction code selection.

Research Interests

Quantum compiler optimization and error correction code selection; fault-tolerant quantum circuit design; LLVM/MLIR-based compiler backends for quantum workloads; reproducible benchmarking and structured experimental design; quantum-classical hybrid methods.

Education

Rochester Institute of Technology

Master of Science, Data Science — GPA 3.9

Expected Aug 2026

Rochester, NY

- Concentration: Bioinformatics & Artificial Intelligence.
- Relevant coursework: Foundations of Data Science (DSCI-633), Software Engineering for Data Science (DSCI-644), Applied Data Science I (DSCI-601), Bioinformatics Algorithms (BIOL-630), Data-Driven Knowledge Discovery (ISTE-780).

University of Rochester, Warner School of Education

Master of Science track, Teaching Computer Science K-12 — GPA 4.0

Expected Aug 2026

Rochester, NY

- NSF Robert Noyce Teacher Scholarship Program (\$30,000 full funding).
- Advanced Certificate in Leadership in Disability and Inclusive Practices.

Rochester Institute of Technology

Master of Science, Computer Science — GPA 3.2

2015

Rochester, NY

- Concentrations: Computer Graphics, Artificial Intelligence, Big Data Analytics.
- Coursework in compiler design, algorithm analysis, and distributed systems.

Farmingdale State College

Bachelor of Science, Computer Engineering Technology — GPA 3.3

2012

Farmingdale, NY

- Dual degree: Electrical Engineering & Computer Engineering Technology.
- Honors: CSTEP Award & Merit, Dual BS Scholarship.

Research and Professional Experience

Rochester Institute of Technology

Fall 2025 – Present

Graduate Assistant, Quantum and AI Research

Rochester, NY

- Conduct research on quantum network path optimization and fault-tolerant routing under adversarial and stochastic conditions.
- Develop compiler-level experimental infrastructure for quantum circuit workloads using Qiskit and Cirq.
- Implement and validate quantum error correction code selection workflows in simulation (Stim-compatible pipelines).
- Apply LLVM/MLIR IR-level analysis concepts to quantum compilation pipeline studies; reviewed TAFFO precision-tuning methodology for applicability to quantum resource management.

University of Rochester, Warner School of Education

Sep 2024 – Present

CS Teacher Candidate & Education Research (NSF Noyce Scholar)

Rochester, NY

- Deliver K–12 computer science instruction in active public school placement as part of the NSF Robert Noyce Scholar program, including computational thinking, inclusive pedagogy, and NYS learning standards.
- Conduct applied education research on AI-informed and data-driven approaches to improve learning outcomes for neurodivergent students, applying Universal Design for Learning (UDL) and evidence-based adaptive strategies.
- Produce lesson artifacts and instructional analyses documenting the intersection of AI, inclusive education, and evidence-based CS pedagogy.

Rochester Institute of Technology

2024 – 2025

Equitable Bioinformatics Research — ISTE-780

Rochester, NY

- Applied systematic fairness analysis to RNA secondary structure prediction pipelines; demonstrated statistical significance ($p < 0.01$) in bias detection across algorithmic variants using SHAP and disparity-impact analysis.
- Developed reproducible evaluation codebase (900+ lines) with automated fairness-metric reporting and statistical testing.

Rochester Institute of Technology

2025 – 2026

Computational Biology Research (BIO614 and BIO550)

Rochester, NY

- Led formal computational analysis in BIO614: advanced RNA secondary structure prediction with thermodynamic MFE enhancements; produced *BIO614-FinalProjectProposal* (Overleaf manuscript).
- Conducted reproducible bulk RNA-seq differential-expression analysis (BIO550): DESeq2 pipeline with QC validation and biological interpretation of nociceptor gene expression.

VEDADATA Inc.

Sep 2022 – Aug 2024

Data Solutions Engineer

New York, NY

- Designed scalable Python workflows for ingestion, validation, and statistical diagnostics, emphasizing data reliability and structured quality checks.
- Strengthened production reliability through anomaly detection, schema validation, and structured logging.

VIOME Inc.

Jan 2021 – Sep 2022

AI Data Solutions Engineer

New York, NY

- Developed healthcare ML workflows for feature engineering, model experimentation, and reliability-aware evaluation in AWS-based production environments.
- Implemented uncertainty-quantification and model-monitoring components to flag prediction instability under distribution shift.

Selected Technical Writing

- *Big Data Medical Diagnosis: A Multi-Method Approach* (RIT M.S. Computer Science graduate research, 2015) — ensemble and deep learning methods for clinical classification on high-dimensional biomedical data.
- *Predicting Hospital Readmission Rates: A Multi-Method ML Analysis* (DSCI-633 Project Report, 2025) — ensemble and regularized regression methods for clinical risk stratification under distribution shift.
- *BIO614-FinalProjectProposal* (Overleaf manuscript, 2026) — RNA secondary structure prediction with thermodynamic deep learning enhancements, reproducible benchmarking, and biological interpretation.
- *Equitable Bioinformatics: Enhancing Diagnostic Decision-Making through RNA and Biomarker Data* (ISTE-780 Project Report, 2025) — fairness-aware ML for genomic diagnostic algorithms with SHAP-based bias detection and statistical significance testing.
- *Differential Gene Expression in Murine DRG Neurons Following Sciatic Nerve Injury: An NGS Reanalysis* (BIOL550 Computational Biology Project, 2026) — reproducible bulk RNA-seq pipeline using DESeq2, QC validation, and pathway-level biological interpretation of nociceptor gene expression.
- *Fairness-Aware Bandits for Clinical Decision Systems* (DSCI-601 Project Proposal, 2025) — bandit learning with equity constraints applied to high-stakes clinical diagnostic decision-making, addressing demographic disparity in sequential clinical recommendation workflows.
- *Quantum Network Path Optimization and Fault-Tolerant Routing* (manuscript under review, 2025–2026) — quantum-classical hybrid ML for fault-tolerant routing under noise constraints and decoherence.

Awards & Recognition

- NSF Robert Noyce Teacher Scholarship (2024–2026) — full funding for M.S. in Teaching Computer Science K–12.
- MS International Scholarship (2012–2015) — MESCYT & RIT, awarded for academic excellence.
- Honorable Mention Technology Award (2012) — NYS STEP/CSTEP, 2nd place capstone project.
- IEEE Alumni Member (2009–Present) — Member No. 90613000.

Technical Competencies

Programming: Python, R, Java, C++, C#, MATLAB, SQL, JavaScript, Bash

Quantum Computing: Qiskit, Cirq (gate-based quantum circuit simulation and algorithm development)

Compiler & Systems: LLVM/MLIR compiler backend framework; TAFFO/Precimonious (approximate computing and precision tuning)

Machine Learning: PyTorch, TensorFlow, scikit-learn, XGBoost, fairness-aware ML, contextual bandit workflows

Data & Analysis: pandas, NumPy, SciPy, SHAP analysis, statistical testing, reproducible benchmarking

Infrastructure: AWS, GCP, Docker, Git/GitHub, Linux, LaTeX

Selected Fit for NTNU Quantum Compiler Technologies Position

- Directly relevant research in quantum circuit optimization and fault-tolerant routing; active study of compiler-assisted precision tuning methodologies (TAFFO, Precimonious) and their extension to QEC code selection.
- Hands-on experience with LLVM/MLIR IR-level analysis and quantum circuit frameworks (Qiskit, Cirq) providing a practical foundation for compiler backend contributions within the SPQR project.
- Reproducibility-first experimental discipline (multi-environment: local, Colab, GCP) with traceable infrastructure—well matched to SSG’s open-source and benchmark-driven research culture (CGO, PLDI).
- Interest in PoliMi collaboration; familiar with TAFFO’s Value Range Analysis approach as the classical precedent the SPQR project extends to the quantum domain.